TEAM NAME: VECTOR CRAFT

TOPIC: POWER LINE AND TOWER INSPECTION DRONE

DAY 1 UPDATE

- Finalised Hexacopter as sudden wind gusts are mentioned in the problem statement. 6 motors share the correction work. So, each one only makes small thrust corrections. And also, it's less complicated than octa copter.
- Payload: 1.5 kg approx. = 25-35% of all up weight. The AUW is 4.5 to 6 kg. TWR = 2(for stability purpose); so total thrust is 12kgs. Thrust per motor is 2kg.
- Diameter of the propeller = 460mm. A larger propeller disc area distributes the 6 kg all-up weight (AUW) reducing the thrust required per individual motor and lowering power consumption during hover. Larger blades (18 inches) operate at lower RPMs compared to smaller props, which significantly decreases aerodynamic drag. stability against 12 m/s wind gusts.
- P_ideal = 289W. we apply a typical Figure of Merit (FOM) for multirotors, which generally ranges from 0.60 to 0.70. so, power now is 445W = 289/0.7
- 6S3P lithium-ion battery pack configuration. Nominal Voltage (22.2 V)
- 44.9 mins flight time (total energy of battery/hover power)
- Flight controller: Pixhawk class
- Payload: LIDAR Thermal IR camera
- Navigation sensor: GPS/GNSS module; Optical sensor
- Total hover current = 20A; Total max current = 43A
- 6 ESCs (30 A)
- Current and voltage sensors for power monitoring

PAYLOAD SENSOR PLACEMENT:

- Mounting & Vibration Damping: We will suspend the 1.5 kg LiDAR and thermal IR camera beneath the main frame using a custom carbon-fiber mounting plate equipped with custom rubber vibration dampers to isolate high-frequency motor oscillations and ensure clean imaging data.

- Stand-Off Distance Management: We will position the optical payload with a slight forward-downward tilt to guarantee the required 3 m stand-off distance from high-voltage towers while maintaining a clear, unobstructed field of view away from the landing gear and propeller wash.
- Wiring & EMI Protection: The sensor data lines and power cables are routed through shielded tubes away from the high-current power distribution board to prevent electromagnetic interference from disrupting the telemetry and navigation sensors.
- Accessibility: Designing the pod assembly with a quick-release slide-in mechanism so maintenance or sensor swaps can be performed rapidly between flight checks.

NAVIGATION SENSOR PLACEMENT:

- GPS / GNSS Module: We mounted our GPS and compass module on an elevated carbon-fiber mast above the top plate. Elevating the unit isolates its internal compass from high-current power traces, ESCs, and battery leads, which protects our navigation from high-voltage EMI near live transmission lines.
- Inertial Measurement Unit (IMU): We placed the IMU directly within our flight controller, positioned at the exact center of gravity (CG) of the frame. Mounting the flight controller on dampening pads minimizes high-frequency motor vibrations and avoids rotational lever-arm errors during rapid maneuvers.
- Optical Sensor: We positioned our optical flow sensor on the bottom plate facing straight down with a completely unobstructed view. Keeping it close to the vertical axis gives us stable, low-altitude position holding and smooth hover adjustments during close-up inspection passes.

FAIL-SAFE PROTOCOL:

- Loss of Link (RC / Telemetry) & High-Voltage Proximity: f we lose our control link or telemetry signal near live conductors, our Pixhawk flight controller automatically initiates a failsafe protocol. Instead of a standard blind Return-to-Launch (RTL) which could drift into transmission towers, the drone triggers a localized Hold-and-Land or a pre-calculated vector retreat away from the structure.

- GPS Signal Degradation / Compass Interference: High-voltage lines can cause severe magnetic interference or GPS multipath errors. To mitigate this, our avionics switch autonomously from GPS to optical flow and altitude hold if compass or GPS health drops below safe thresholds, preventing sudden flyaways.
- Motor-Out Redundancy: Because we chose a 6-motor (hexacopter) configuration, the system can safely compensate if a single motor or ESC fails. The flight controller automatically redistributes thrust to the remaining five motors, allowing us to maintain stable attitude control and execute a controlled emergency landing.

REGARDING THE EMI INTERFERENCE PROBLEMS: (solutions)

- Separating GNS and compass (reduces the interference from onboard electronic components)
- Organise power and signal wiring (keep GPS cables and ESC cables away from each other)
- Appropriate shielding cables and filtering
- Using optical flow sensors to not entirely depend upon GPS
- Adding failsafe
- Using CANbus as communication protocol (CAN uses differential signalling, which helps reject noise coupled similarly into both signal wires.)

1. Completed Assemblies & Modeling Work

- **Chassis Core Stack:** Designed a 3-plate central hub structure (340 mm $\times$ 260 mm plates with 20 mm corner fillets) featuring 92 mm lower and 70 mm upper standoff bays. Integrated Ø4.5 mm plate clearance holes aligned over Ø3.3 mm internal standoff threads, along with dual Ø30 mm cable pass-throughs on the middle plate.
- **Modular 6-Arm Array:** Modeled the master arm assembly along the X-axis including aluminum clamp blocks ($50 \times 44 \times 40$ mm), inboard CFRP tubes (Ø28/Ø25 mm), aluminum sleeve joints (Ø34/Ø28 mm), and outboard GFRP tubes (Ø28/Ø25 mm). Circularly patterned all 6 arms radially at $60°$ intervals.

- **Propulsion Units & Rigid Kinematic Joints:** Integrated 18-inch counter-rotating propeller pairs ($3 \times$ CW and $3 \times$ CCW) onto $\emptyset 60 \times 35$ mm motor placeholders using zero-degree-of-freedom rigid joints. Seated prop hubs flush onto the motor shaft studs ($Z = 128$ mm).

2. Technical Parameters & Geometric Specifications

- **Chassis Footprint / Motor-to-Motor Distance:** 1120 mm rotor disc wheelbase (560 mm radial distance from frame center to motor shaft axes).
- **Propeller Tip Clearance:** ≈ 52 mm spacing between adjacent 18-inch propeller blade tips, preventing inter-blade tip vortex interference during high thrust operations.
- **Assembly Datum & Axes:** Z-up global coordinate system with frame origin $(0,0,0)$ set at the geometric center of the bottom plate.

3. Core Technical Justifications

- **Low Reynolds Airfoil & Twist Distribution:** Selected a cambered, low-Reynolds propeller profile with geometric washout (flatter pitch at tips) to balance induced drag, prevent tip stall, and optimize hover efficiency (445 W power draw at 6 kg total thrust).
- **Torque Neutralization:** Paired $3 \times$ CW and $3 \times$ CCW rotation sets to achieve net-zero reactive torque on the airframe during hover and enable precise yaw control via RPM modulation.
- **Structural Redundancy:** Configured as a hexacopter layout to maintain flight stability and execute safe controlled landings in the event of a single-motor failure.